THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** 140 degrees	Initial i = 55; new i = 70; angle = 140.
3	**(b)** 90 degrees to the surface (back along normal)	Incidence along the normal.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** 65 degrees	i2 = 90 - 25.
6	**(a)** parallel	Parallel beam.
7	**(c)** A is true but R is false.	A is true; R is false because they form virtual diminished images.
8	**(c)** 8 m	Boy is 5 m from the mirror; image 5 m behind; man at 3 m; total 8.
9	**(c)** beyond C	Beyond C.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Initial i = 60 degrees (1); New i = 70 or 50 degrees (1).	1+1
12	Size depends on reflection geometry, not distance (1); Image is as big as the object (1).	1+1
13	It bends parallel rays towards each other (1); They meet at the principal focus (1).	1+1
14	Image moves at 3 m/s away from the mirror (1); Relative speed = 6 m/s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	n = 360/theta - 1 = 7 (1); 360/theta = 8 (1); theta = 45 degrees (1).	1+1+1
16	At first the object is within F (1); Beyond F image is real and inverted (1); Hence face appears inverted (1).	1+1+1
17	Both use convex lenses (1); Camera uses a film or sensor; eye uses retina (1); Both give real inverted images (1).	1+1+1
	OR: Image is 15 cm behind the mirror (1); Distance from object = 30 cm (1); New: object at 11 cm, so distance = 22 cm (1).	1+1+1
18	Concave reflector with bulb at F (1); Reflected rays become parallel (1); Gives a strong narrow beam (1).	1+1+1
	OR: Convex: virtual, erect (1); Always smaller (1); Image lies behind the mirror (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Beyond C: real, inverted, diminished (1); Mirror has R = 20 cm so C = 20 cm (1); 30 cm is beyond C (1); Image between F and C (1); Real, inverted, smaller (1).	1+1+1+1+1
	OR: Cornea and lens focus light (1); Image forms on retina (1); Myopia: concave lens (1); Hypermetropia: convex lens (1); Cataract and other causes (1).	1+1+1+1+1
20	First mirror: i = r = 35 degrees; the ray makes 55 degrees with the mirror surface (1); In the triangle with the right angle, the ray makes 35 degrees with the second mirror surface (1); Angle of incidence at the second mirror = 55 degrees (1); The emergent ray turns through 180 degrees in all (1); Hence it is parallel to the incident ray but opposite in direction (1).	1+1+1+1+1
	OR: Mirrors need regular reflection (1); Screens use diffuse reflection to be seen from all angles (1); Road signs use retro-reflection (1); Objects are seen by diffuse reflection (1); Both obey the laws (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1:50	12:00 - 10:10.
2	**(c)** smaller and closer to F	Diminished virtual image nears F.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(c)** C to infinity	Object near F gives distant image.
5	**(a)** its optical centre	Central point.
6	**(c)** A is true but R is false.	A is true; R is false: it diverges rays.
7	**(b)** between C and F	Enlarged real image.
8	**(d)** bends rays more strongly	Short f means high power.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Diffuse reflection.
10	**(d)** form an image of the same type but dimmer	Each half still converges.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Convex lens (1); Real if object beyond f; virtual if within f (1).	1+1
12	Lateral inversion (1); Left and right interchange (1).	1+1
13	To fit in the mouth (1); To form enlarged erect images (1).	1+1
14	Gives a wide view (1); Image is erect and diminished (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mirrors are parallel at 45 degrees (1); Each reflection keeps the image same size (1); Two reflections keep it erect (1).	1+1+1
16	Convex forms a bright spot on a screen (1); Concave cannot form a real spot (1); Concave spreads light (1).	1+1+1
17	The ray lies along the normal (1); Angle of incidence zero (1); Angle of reflection zero (1).	1+1+1
	OR: Both form real, inverted, diminished images (1); Mirror: same side as object (1); Lens: opposite side (1).	1+1+1
18	At 25 cm (1); Real (1); Inverted and very small (1).	1+1+1
	OR: Place the mirror on the ground (1); Move until the top of the tree is seen (1); Similar triangles give the height (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Telescope: objective lens or mirror collects light (1); Microscope: two convex lenses (1); Camera: convex lens, real image (1); Projector: convex lens, enlarged real image (1); Images on screen are inverted (1).	1+1+1+1+1
	OR: Ray parallel to axis passes through F after reflection (1); Ray from the object to the pole reflects at equal angle (1); Reflected rays diverge (1); Virtual, erect, enlarged image behind mirror (1); Barbers get an enlarged face image (1).	1+1+1+1+1
20	Left-right reversal (1); MAHESH looks reversed (1); 2018 looks reversed (1); 4:30 appears as 7:30 (1); Clock face reversed (1).	1+1+1+1+1
	OR: Place object between mirrors (1); Vary angle and count images (1); n = 360/theta - 1 (1); Table: 11, 7, 5, 3, 2 (1); Smaller angle, more images (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** reflect headlight beams back to the driver	Retroreflector.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(b)** less than 20 cm	Within the focal length.
4	**(d)** a wide view with a smaller erect image	Concave lens.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 15 cm	Sun is at infinity.
7	**(c)** near point	Least distance of distinct vision.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(d)** 4 cm	Pencil at 2 cm; image 2 cm behind.
10	**(d)** too long	Myopia: eyeball elongated.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Lens bends all colours nearly equally (1); Prism disperses light into colours (1).	1+1
12	Object at the centre of curvature C (1); Image forms at C, inverted, real, same size (1).	1+1
13	R = 15 cm (1); 7.5 cm from pole (1).	1+1
14	More images are formed (1); n = 360/theta - 1 increases (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Object far: image close to F, small (1); Object nearer F: image moves farther, larger (1); Magnification depends on distances (1).	1+1+1
16	Hold one mirror behind her head (1); View its image in the second mirror in front (1); Light reflects twice (1).	1+1+1
17	Mirror in front inverts (1); Driver sees them in the rear-view mirror (1); Appear correct (1).	1+1+1
	OR: Object within focal length (1); Rays diverge after the lens (1); Virtual, erect, enlarged (1).	1+1+1
18	Front glass surface reflects weakly (1); Silvered back reflects strongly (1); Faint second images appear (1).	1+1+1
	OR: Eye lens is convex and flexible (1); For near, it becomes thicker (1); For far, thinner (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dentist: concave, enlarged erect image (1); Headlight: concave, parallel beam (1); Shop mirror: convex, wide view (1); Convex image is smaller (1); Choice depends on required effect (1).	1+1+1+1+1
	OR: Light enters through the cornea and pupil (1); Convex eye lens focuses it (1); Real, inverted, diminished image forms on the retina (1); Optic nerve carries the signal to the brain (1); Brain interprets the image as erect (1).	1+1+1+1+1
20	Beyond 2F: smaller image (1); At 2F: same size (1); Between F and 2F: larger image (1); Measure distances and heights (1); Observe image on the screen (1).	1+1+1+1+1
	OR: Fibre optics (total reflection) (1); Endoscopes use lenses (1); Eyeglasses (1); Solar concentrators (1); Laser surgeries and cameras (1).	1+1+1+1+1

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